

```
In [61]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [62]: df = pd.read_csv("netflix_data.csv")
df
```

Out[62]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021
...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody	United States	November 1, 2019	2009

show_id	type	title	director	cast	country	date_added	release_year	
8805	s8806	Movie	Zoom	Peter Hewitt	Harrelson, Emma Stone, ...	United States	January 11, 2020	2006
					Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...			
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015

8807 rows × 12 columns

```
In [63]: df.head()
```

Out[63]:

	show_id	type	title	director	cast	country	date_added	release_year	rating
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	T M
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	T M
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	T M
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	T M

In [64]: df.shape

Out[64]: (8807, 12)

In [65]: df.columns

Out[65]: Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added', 'release_year', 'rating', 'duration', 'listed_in', 'description'], dtype='object')

In [66]: df.info()

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   show_id         8807 non-null   object
 1   type            8807 non-null   object
 2   title           8807 non-null   object
 3   director        6173 non-null   object
 4   cast            7982 non-null   object
 5   country         7976 non-null   object
 6   date_added      8797 non-null   object
 7   release_year    8807 non-null   int64
 8   rating          8803 non-null   object
 9   duration        8804 non-null   object
10   listed_in       8807 non-null   object
11   description      8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB

```

In [67]: `df.describe()`

Out[67]:

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

In [68]: `df.isnull().sum().sort_values(ascending=False)`

Out[68]: **0**

director	2634
country	831
cast	825
date_added	10
rating	4
duration	3
show_id	0
type	0
title	0
release_year	0
listed_in	0
description	0

dtype: int64In [69]: `round(df.isnull().sum()/df.shape[0]*100,2).sort_values(ascending=False)`Out[69]: **0**

director	29.91
country	9.44
cast	9.37
date_added	0.11
rating	0.05
duration	0.03
show_id	0.00
type	0.00
title	0.00
release_year	0.00
listed_in	0.00
description	0.00

dtype: float64

```
In [70]: df["director"].value_counts()
```

```
Out[70]:
```

	count
director	
Rajiv Chilaka	19
Raúl Campos, Jan Suter	18
Suhas Kadav	16
Marcus Raboy	16
Jay Karas	14
...	...
James Brown	1
Ivona Juka	1
Mu Chu	1
Chandra Prakash Dwivedi	1
Majid Al Ansari	1

4528 rows × 1 columns

dtype: int64

```
In [71]: df.type.value_counts()
```

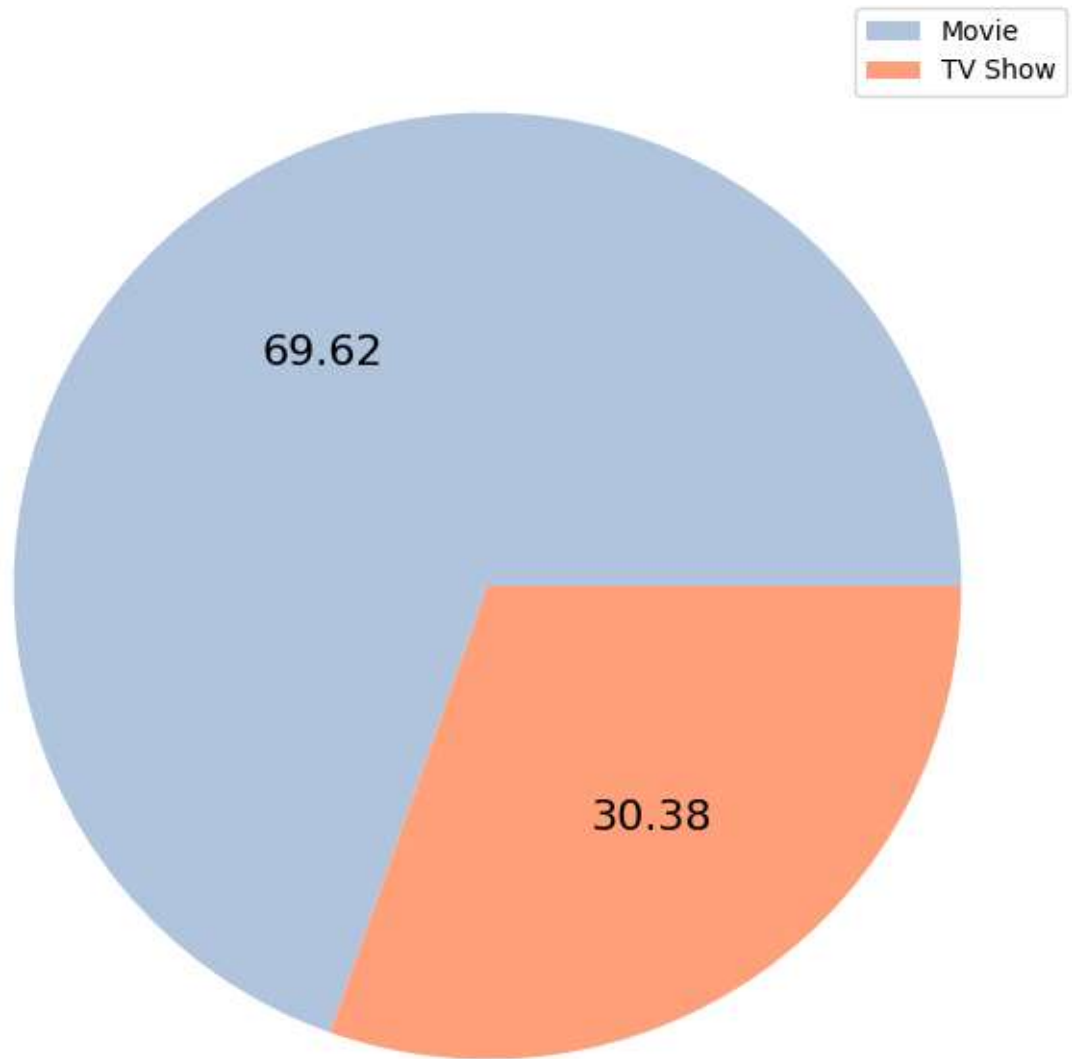
```
Out[71]:
```

	count
type	
Movie	6131
TV Show	2676

dtype: int64

```
In [72]: plt.figure(figsize=(10,8))

plt.pie(df.type.value_counts(),
        labels = df.type.value_counts().index,
        labeldistance = None, autopct="%.2f",
        textprops = {'fontsize': 16,},
        colors = ['lightsteelblue', 'lightsalmon' ] )
plt.legend()
plt.show()
```



```
In [73]: df.rating.value_counts()
```

Out[73]:

rating	count
TV-MA	3207
TV-14	2160
TV-PG	863
R	799
PG-13	490
TV-Y7	334
TV-Y	307
PG	287
TV-G	220
NR	80
G	41
TV-Y7-FV	6
NC-17	3
UR	3
74 min	1
84 min	1
66 min	1

dtype: int64

```
In [74]: df.country.value_counts()
```

Out[74]:

country		count
United States		2818
India		972
United Kingdom		419
Japan		245
South Korea		199
...		...
Mexico, United States, Spain, Colombia		1
Canada, Norway		1
Finland, Germany, Belgium		1
Argentina, United States, Mexico		1
United Kingdom, United States, Germany, Denmark, Belgium, Japan		1

748 rows × 1 columns

dtype: int64

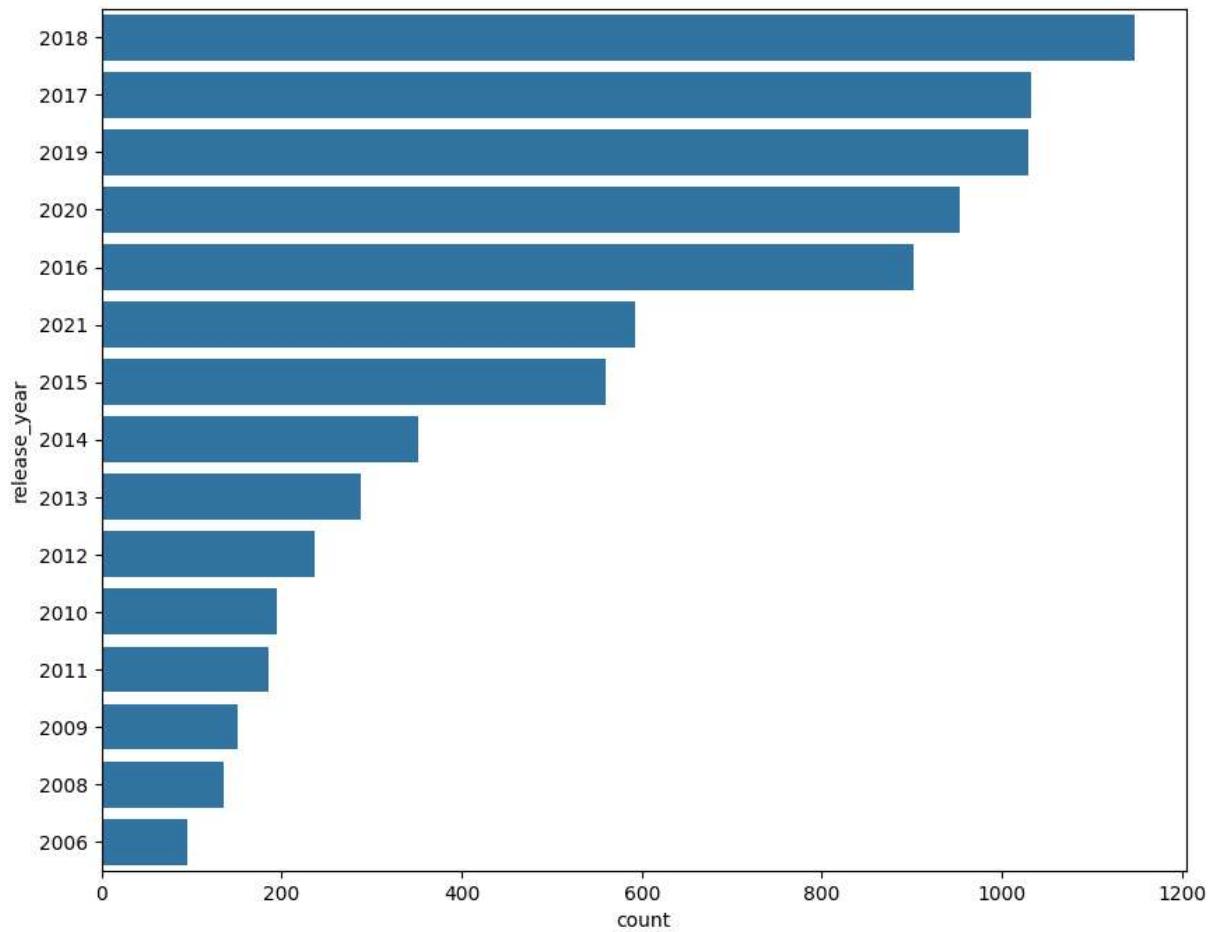
```
In [75]: df.country.value_counts().head(10)
```

Out[75]:

country		count
United States		2818
India		972
United Kingdom		419
Japan		245
South Korea		199
Canada		181
Spain		145
France		124
Mexico		110
Egypt		106

dtype: int64

```
In [76]: plt.figure(figsize=(10,8))  
ax = sns.countplot(y="release_year", data=df, order=df.release_year.value_counts().
```



```
In [77]: df.director.value_counts().head(10)
```

Out[77]:

	count
director	
Rajiv Chilaka	19
Raúl Campos, Jan Suter	18
Suhas Kadav	16
Marcus Raboy	16
Jay Karas	14
Cathy Garcia-Molina	13
Martin Scorsese	12
Youssef Chahine	12
Jay Chapman	12
Steven Spielberg	11

dtype: int64

In [78]: df.listed_in.value_counts().head(10)

Out[78]:

	count
listed_in	
Dramas, International Movies	362
Documentaries	359
Stand-Up Comedy	334
Comedies, Dramas, International Movies	274
Dramas, Independent Movies, International Movies	252
Kids' TV	220
Children & Family Movies	215
Children & Family Movies, Comedies	201
Documentaries, International Movies	186
Dramas, International Movies, Romantic Movies	180

dtype: int64

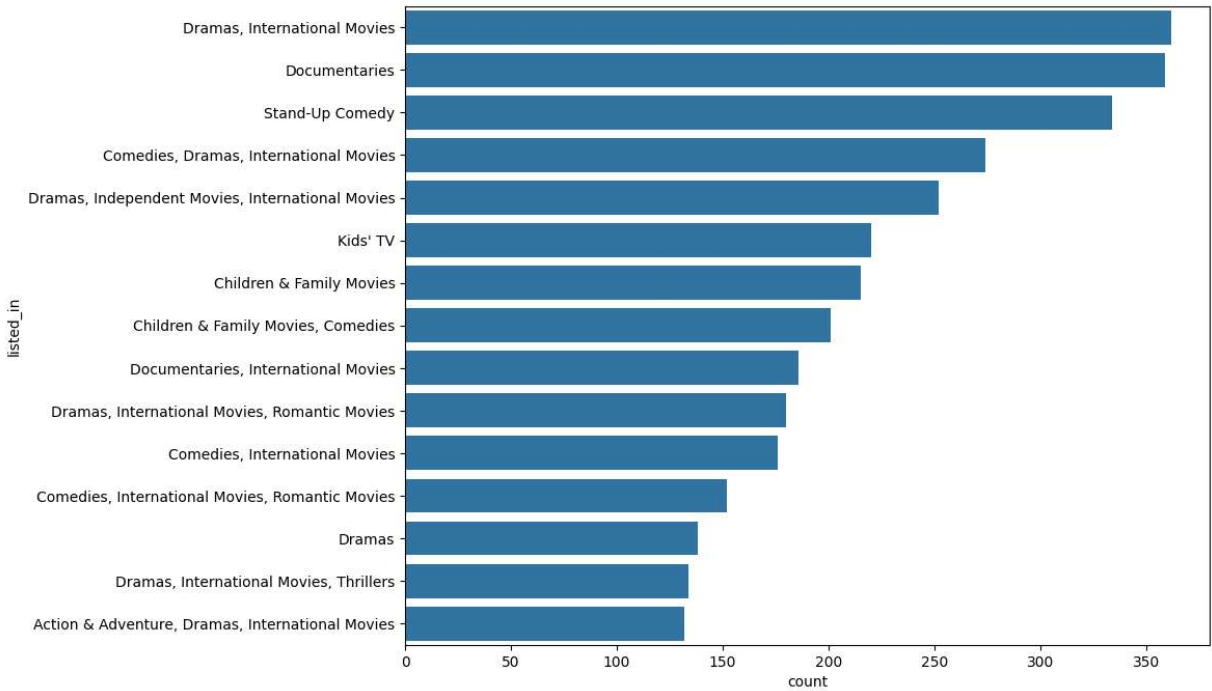
In [79]: df.listed_in.value_counts().tail()

Out[79]:

listed_in	count
Action & Adventure, Cult Movies	1
Action & Adventure, Comedies, Music & Musicals	1
Classic Movies, Horror Movies, Thrillers	1
Children & Family Movies, Classic Movies, Dramas	1
Cult Movies, Dramas, Thrillers	1

dtype: int64

```
In [80]: plt.figure(figsize=(10,8))
ax = sns.countplot(y="listed_in", data=df, order=df.listed_in.value_counts().index[
```



```
In [81]: round(df.isnull().sum()/df.shape[0]*100,2).sort_values(ascending=False)
```

Out[81]: 0

director	29.91
country	9.44
cast	9.37
date_added	0.11
rating	0.05
duration	0.03
show_id	0.00
type	0.00
title	0.00
release_year	0.00
listed_in	0.00
description	0.00

dtype: float64

```
In [82]: round(df.isnull().sum())
```

Out[82]: 0

show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

```
In [83]: df.dropna(subset=["rating", "duration"], axis=0, inplace=True)
```

```
In [84]: df.shape
```

```
Out[84]: (8800, 12)
```

```
In [85]: round(df.isnull().sum()/df.shape[0]*100,2).sort_values(ascending=False)
```

```
Out[85]:
```

	0
--	---

director	29.90
country	9.43
cast	9.38
date_added	0.11
title	0.00
type	0.00
show_id	0.00
release_year	0.00
rating	0.00
duration	0.00
listed_in	0.00
description	0.00

dtype: float64

```
In [86]: df.dropna(subset=["date_added"], axis=0, inplace=True)
```

```
In [87]: round(df.isnull().sum()/df.shape[0]*100,2).sort_values(ascending=False)
```

Out[87]:

0

director	29.82
country	9.43
cast	9.39
show_id	0.00
title	0.00
type	0.00
date_added	0.00
release_year	0.00
rating	0.00
duration	0.00
listed_in	0.00
description	0.00

dtype: float64In [88]: `df["country"].replace(np.nan, "Unknown", inplace=True)`

/tmp/ipykernel_2979/1299151209.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df["country"].replace(np.nan, "Unknown", inplace=True)
```

In [89]: `round(df.isnull().sum()/df.shape[0]*100,2).sort_values(ascending=False)`

Out[89]:

	0
director	29.82
cast	9.39
type	0.00
show_id	0.00
title	0.00
country	0.00
date_added	0.00
release_year	0.00
rating	0.00
duration	0.00
listed_in	0.00
description	0.00

dtype: float64

```
In [90]: df.country.value_counts().head()
```

Out[90]:

	count
country	
United States	2809
India	972
Unknown	829
United Kingdom	418
Japan	243

dtype: int64

```
In [91]: df.cast.value_counts().head(10)
```

Out[91]:

	count
cast	
David Attenborough	19
Vatsal Dubey, Julie Tejwani, Rupa Bhimani, Jigna Bhardwaj, Rajesh Kava, Mousam, Swapnil	14
Samuel West	10
Jeff Dunham	7
Michela Luci, Jamie Watson, Eric Peterson, Anna Claire Bartlam, Nicolas Aqui, Cory Doran, Julie Lemieux, Derek McGrath	6
Craig Sechler	6
David Spade, London Hughes, Fortune Feimster	6
Kevin Hart	6
Iliza Shlesinger	5
Bill Burr	5

dtype: int64In [92]: `df["cast"].replace(np.nan, "No Cast", inplace=True)`

/tmp/ipykernel_2979/2806194272.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df["cast"].replace(np.nan, "No Cast", inplace=True)
```

In [93]: `df["director"].replace(np.nan, "No Director", inplace=True)`

/tmp/ipykernel_2979/1963959508.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df["director"].replace(np.nan, "No Director", inplace=True)
```

```
In [94]: round(df.isnull().sum()/df.shape[0]*100,2).sort_values(ascending=False)
```

```
Out[94]:
```

	0
--	---

show_id	0.0
type	0.0
title	0.0
director	0.0
cast	0.0
country	0.0
date_added	0.0
release_year	0.0
rating	0.0
duration	0.0
listed_in	0.0
description	0.0

dtype: float64

```
In [95]: df["title"]
```

Out[95]:

	title
0	Dick Johnson Is Dead
1	Blood & Water
2	Ganglands
3	Jailbirds New Orleans
4	Kota Factory
...	...
8802	Zodiac
8803	Zombie Dumb
8804	Zombieland
8805	Zoom
8806	Zubaan

8790 rows × 1 columns

dtype: object

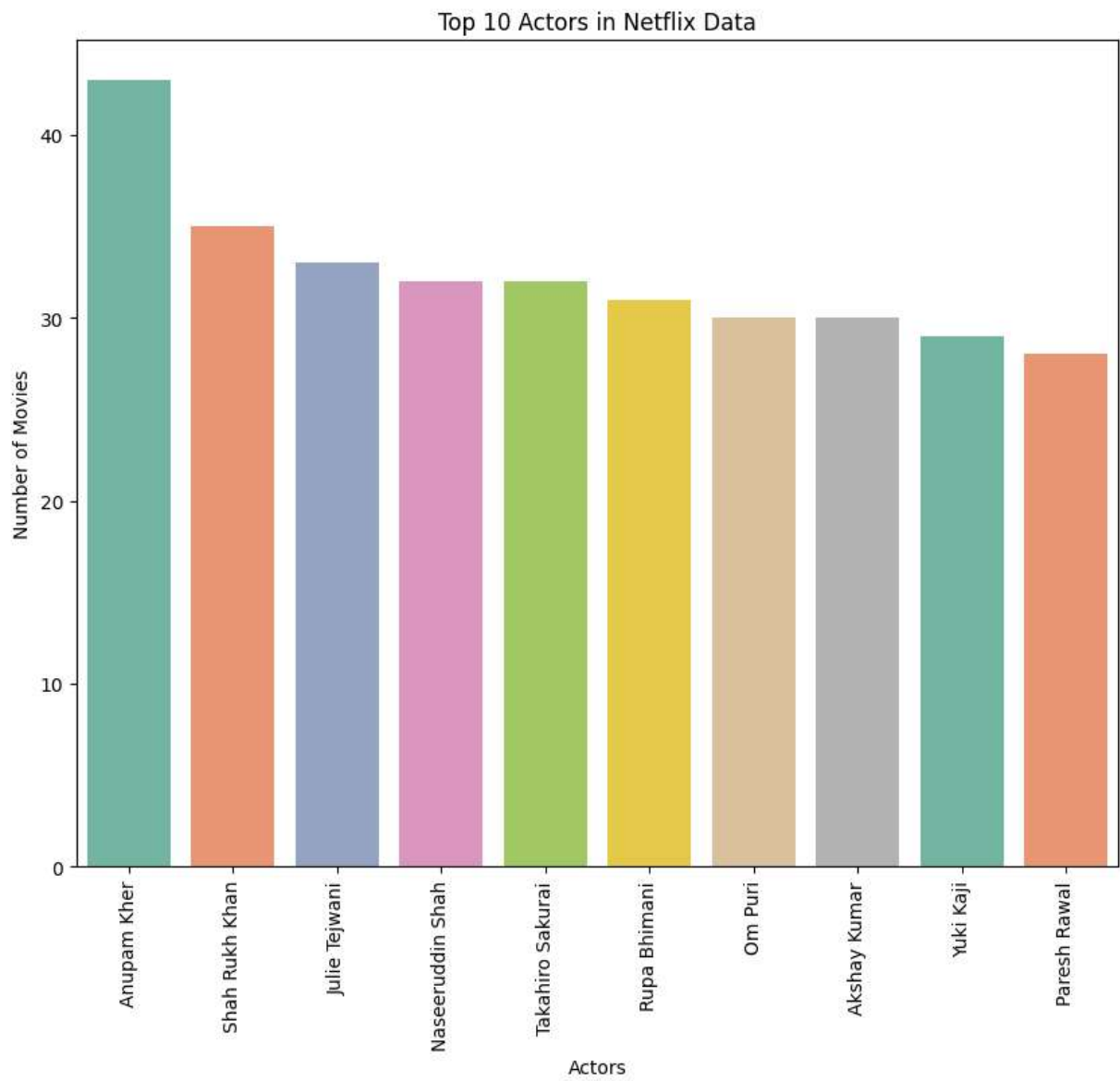
```
In [96]: cast_shows = df[df["cast"] != "No Cast"].set_index("title").cast.str.split(", ", ex
top_10_actors = cast_shows.value_counts().head(10)

plt.figure(figsize=(10, 8))
sns.barplot(x=top_10_actors.index, y=top_10_actors.values, palette='Set2')
plt.xticks(rotation=90)
plt.xlabel("Actors")
plt.ylabel("Number of Movies")
plt.title("Top 10 Actors in Netflix Data")
plt.show()
```

/tmp/ipykernel_2979/2800895399.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.barplot(x=top_10_actors.index, y=top_10_actors.values, palette='Set2')
```



```
In [97]: movies_df = df.loc[(df["type"] == "Movie")]
movies_df.head()
```

Out[97]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	No Cast	United States	September 25, 2021	2021
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	Unknown	September 24, 2021	2021
7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D... D...	United States, Ghana, Burkina Faso, United Kin...	September 24, 2021	1990
9	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	September 24, 2021	2020
12	s13	Movie	Je Suis Karl	Christian Schwochow	Luna Wedler, Jannis Niewöhner, Milan Peschel, ...	Germany, Czech Republic	September 23, 2021	2020

```
In [98]: show_df = df.loc[(df["type"] == "TV Show")]
show_df.head()
```

Out[98]:

	show_id	type	title	director	cast	country	date_added	release_year	rat
1	s2	TV Show	Blood & Water	No Director	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	September 24, 2021	2021	
3	s4	TV Show	Jailbirds New Orleans	No Director	No Cast	Unknown	September 24, 2021	2021	
4	s5	TV Show	Kota Factory	No Director	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	
5	s6	TV Show	Midnight Mass	Mike Flanagan	Kate Siegel, Zach Gilford, Hamish Linklater, H...	Unknown	September 24, 2021	2021	



In [99]: `movies_df.duration = movies_df.duration.apply(lambda x: x.replace(" min", "") if 'm' in x else x)`
`movies_df.head()`

/tmp/ipykernel_2979/3664425586.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using `.loc[row_indexer,col_indexer] = value` instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
movies_df.duration = movies_df.duration.apply(lambda x: x.replace(" min", "") if 'm' in x else x)
```

Out[99]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	No Cast	United States	September 25, 2021	2021
6	s7	Movie	My Little Pony: A New Generation	Robert Cullen, José Luis Ucha	Vanessa Hudgens, Kimiko Glenn, James Marsden, ...	Unknown	September 24, 2021	2021
7	s8	Movie	Sankofa	Haile Gerima	Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...	United States, Ghana, Burkina Faso, United Kin...	September 24, 2021	1990
9	s10	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd, Kevin Kline, T...	United States	September 24, 2021	2020
12	s13	Movie	Je Suis Karl	Christian Schwochow	Luna Wedler, Jannis Niewöhner, Milan Peschel, ...	Germany, Czech Republic	September 23, 2021	2020

In [100...

```
movies_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 6126 entries, 0 to 8806
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   show_id         6126 non-null   object
1   type            6126 non-null   object
2   title           6126 non-null   object
3   director        6126 non-null   object
4   cast            6126 non-null   object
5   country         6126 non-null   object
6   date_added      6126 non-null   object
7   release_year    6126 non-null   int64
8   rating          6126 non-null   object
9   duration        6126 non-null   object
10  listed_in       6126 non-null   object
11  description     6126 non-null   object
dtypes: int64(1), object(11)
memory usage: 751.2+ KB
```

In [101...

movies_df.loc[:,["duration"]] = movies_df.loc[:,["duration"]].apply(lambda x: x.astype(int))
movies_df.describe()

Out[101...

release_year	
count	6126.000000
mean	2013.120144
std	9.681723
min	1942.000000
25%	2012.000000
50%	2016.000000
75%	2018.000000
max	2021.000000

In [102...

shortest_movies = movies_df.loc[(movies_df['duration']==np.min(movies_df.duration))]
shortest_movies

Out[102...

	show_id	type	title	director	cast	country	date_added	release_year	rating
3777	s3778	Movie	Silent	Limbert Fabian, Brandon Oldenburg	No Cast	United States	June 4, 2019	2014	TV-Y



```
In [103...] longest_movies = movies_df.loc[(movies_df['duration']==np.max(movies_df.duration))]
longest_movies
```

```
Out[103...]
```

	show_id	type	title	director	cast	country	date_added	release_year
4253	s4254	Movie	Black Mirror: Bandersnatch	No Director	Fionn Whitehead, Will Poulter, Craig Parkinson...	United States	December 28, 2018	2018

```
In [104...] show_df.duration = show_df.duration.apply(lambda x: x.replace(" Season", "")) if 'Season' in x else x
show_df.head(3)
```

/tmp/ipykernel_2979/554293217.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
show_df.duration = show_df.duration.apply(lambda x: x.replace(" Season", "") if 'Season' in x else x)
```

```
Out[104...]

```

	show_id	type	title	director	cast	country	date_added	release_year	rating
1	s2	TV Show	Blood & Water	No Director	Ama Qamata, Khosi Ngema, Gail Mabalane, Thabane...	South Africa	September 24, 2021	2021	7.5
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	September 24, 2021	2021	7.5
3	s4	TV Show	Jailbirds New Orleans	No Director	No Cast	Unknown	September 24, 2021	2021	7.5

```
In [105...] show_df.duration = show_df.duration.apply(lambda x: x.replace("s", "")) if 's' in x else x
show_df.head(3)
```

```
/tmp/ipykernel_2979/3915305098.py:1: SettingWithCopyWarning:  
A value is trying to be set on a copy of a slice from a DataFrame.  
Try using .loc[row_indexer,col_indexer] = value instead  
  
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/u  
ser_guide/indexing.html#returning-a-view-versus-a-copy  
show_df.duration = show_df.duration.apply(lambda x: x.replace("s", "") if 's' in x  
else x)
```

Out[105...

	show_id	type	title	director	cast	country	date_added	release_year	rating
1	s2	TV Show	Blood & Water	No Director	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-14
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	September 24, 2021	2021	TV-14
3	s4	TV Show	Jailbirds New Orleans	No Director	No Cast	Unknown	September 24, 2021	2021	TV-14



```
In [106... show_df.loc[:,["duration"]] = show_df.loc[:,["duration"]].apply(lambda x: x.astype(  
show_df.describe())
```

Out[106...

	release_year
count	2664.000000
mean	2016.627628
std	5.735194
min	1925.000000
25%	2016.000000
50%	2018.000000
75%	2020.000000
max	2021.000000

```
In [107... show_df.duration.value_counts().tail(10)
```

```
Out[107... count
```

duration	
6	33
7	23
8	17
9	9
10	6
13	2
12	2
15	2
17	1
11	1

dtype: int64

```
In [108... longest_shows = show_df.loc[(show_df["duration"] >= 13)]  
longest_shows
```

Out[108...

	show_id	type	title	director	cast	country	date_added	release_year
548	s549	TV Show	Grey's Anatomy	No Director	Ellen Pompeo, Sandra Oh, Katherine Heigl, Just...	United States	July 3, 2021	2020
1354	s1355	TV Show	Heartland	No Director	Amber Marshall, Michelle Morgan, Graham Wardle...	Canada	February 1, 2021	2019
2423	s2424	TV Show	Supernatural	Phil Sgriccia	Jared Padalecki, Jensen Ackles, Mark Sheppard,...	United States, Canada	June 5, 2020	2019
4220	s4221	TV Show	COMEDIANS of the world	No Director	Neal Brennan, Chris D'Elia, Nicole Byer, Nick ...	United States	January 1, 2019	2019
4798	s4799	TV Show	NCIS	No Director	Mark Harmon, Michael Weatherly, Pauley Perrett...	United States	July 1, 2018	2017

In [109...

```
longest_shows.rating.value_counts()
```

Out[109...

	count
rating	
TV-14	4
TV-MA	1

dtype: int64

Release Year

```
In [110... last_decade = df[["type", "release_year"]]
last_decade = last_decade.rename(columns = {"release_year" : "Release Year"})
last_decade = last_decade[last_decade["Release Year"]>=2010]
last_decade
```

```
Out[110...      type  Release Year
0      Movie      2020
1  TV Show      2021
2  TV Show      2021
3  TV Show      2021
4  TV Show      2021
...      ...      ...
8798  Movie      2014
8800  TV Show      2012
8801  Movie      2015
8803  TV Show      2018
8806  Movie      2015
```

7458 rows × 2 columns

```
In [111... last_decade_df = last_decade.groupby("Release Year")["type"].size().reset_index()
last_decade_df = pd.DataFrame(last_decade_df)
last_decade_df
```

Out[111...

	Release Year	type
0	2010	192
1	2011	185
2	2012	236
3	2013	286
4	2014	352
5	2015	555
6	2016	901
7	2017	1030
8	2018	1146
9	2019	1030
10	2020	953
11	2021	592

In [112...

```
last_decade_df.rename(columns = {"type": "Total Content"}, inplace = True)
```

In [113...

```
last_decade.groupby("Release Year")["type"].value_counts()
```

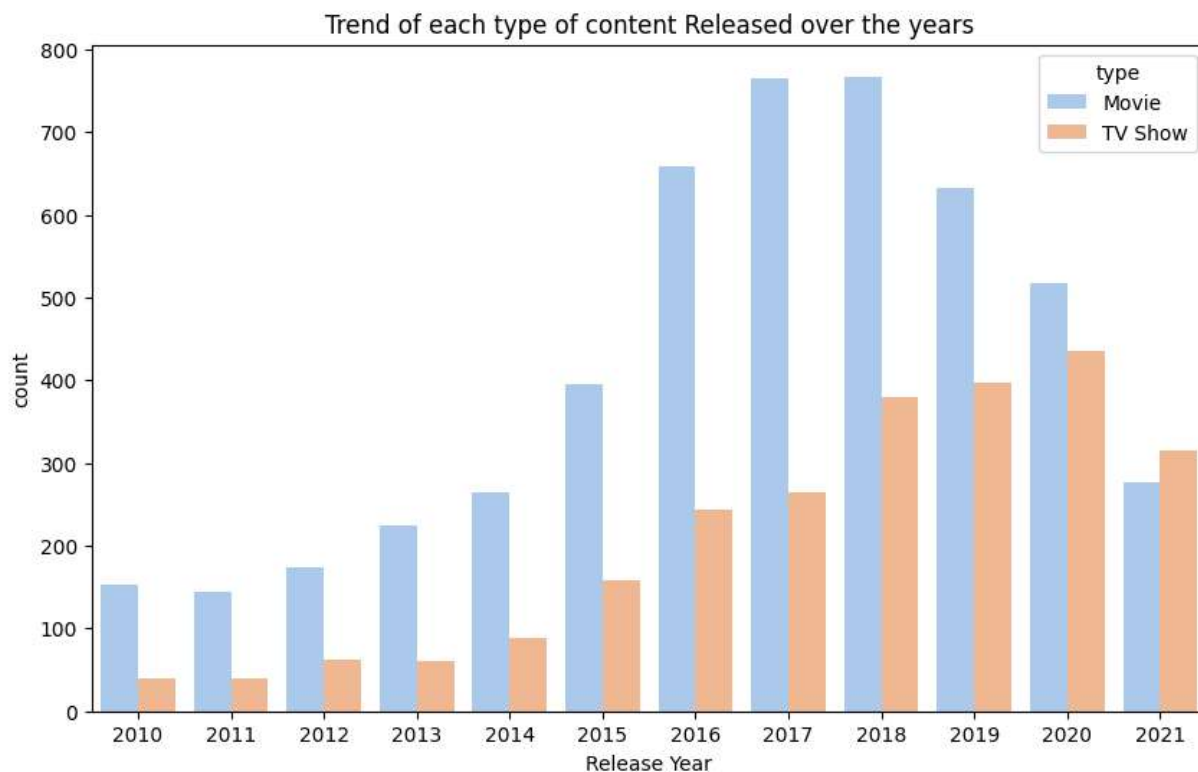
Out[113...

Release Year	type	count
2010	Movie	153
	TV Show	39
2011	Movie	145
	TV Show	40
2012	Movie	173
	TV Show	63
2013	Movie	225
	TV Show	61
2014	Movie	264
	TV Show	88
2015	Movie	396
	TV Show	159
2016	Movie	658
	TV Show	243
2017	Movie	765
	TV Show	265
2018	Movie	767
	TV Show	379
2019	Movie	633
	TV Show	397
2020	Movie	517
	TV Show	436
2021	TV Show	315
	Movie	277

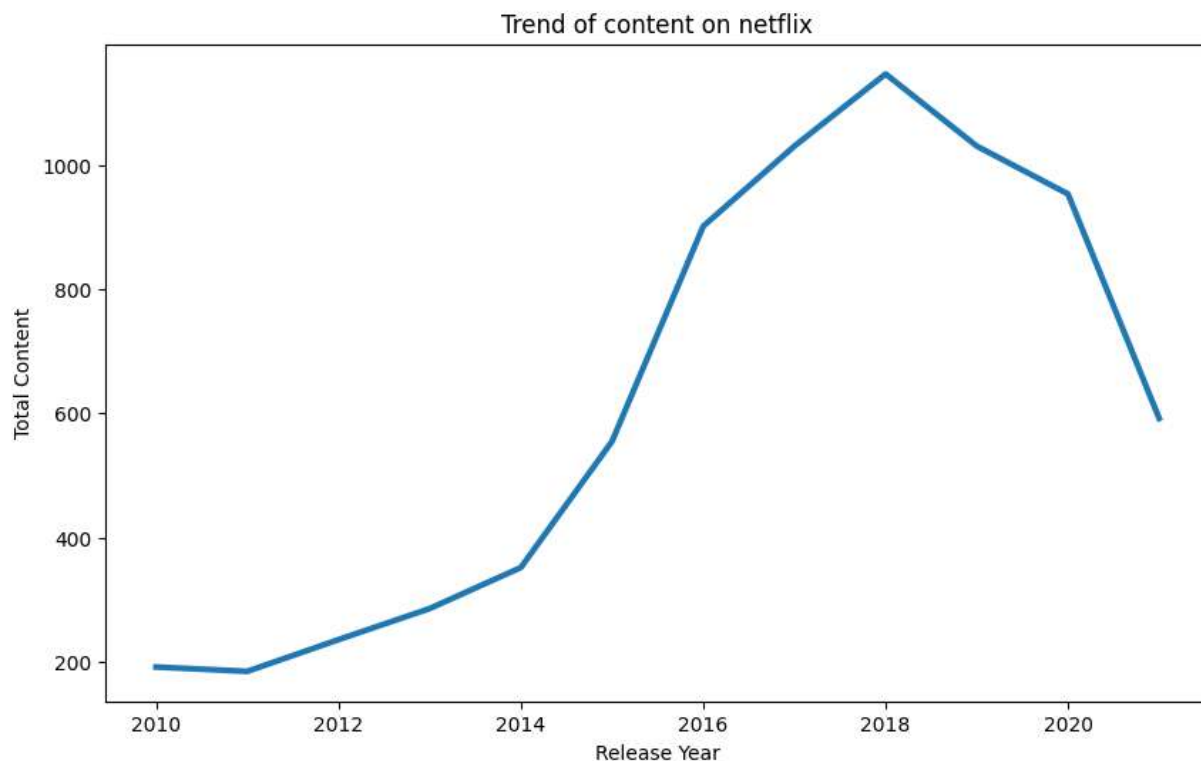
dtype: int64

In [114...

```
plt.figure(figsize = (10,6))
count_plot = sns.countplot(x = "Release Year", data = last_decade, hue="type",
                           palette= "pastel")
count_plot.set(title = "Trend of each type of content Released over the years")
plt.show()
```



```
In [115... plt.figure(figsize = (10,6))
plot_total_content= sns.lineplot(x= "Release Year", y = "Total Content", data = las
                                linewidth = 3)
plot_total_content.set(xlabel = "Release Year", ylabel = "Total Content",
                        title = "Trend of content on netflix")
plt.show()
```



Countries

```
In [116... top_10_countries= df.country.value_counts().head(10)
top_10_countries = pd.DataFrame(top_10_countries)
top_10_countries
```

Out[116...

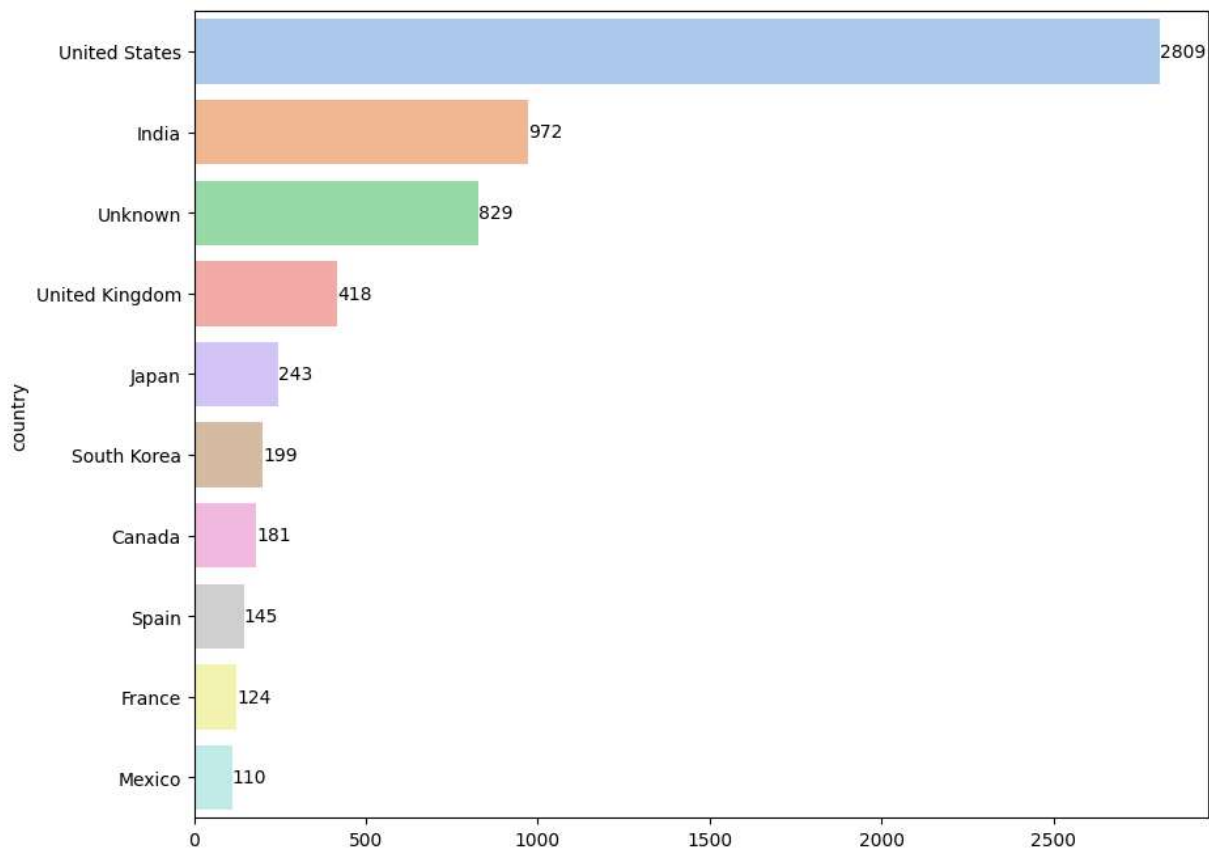
	count
country	
United States	2809
India	972
Unknown	829
United Kingdom	418
Japan	243
South Korea	199
Canada	181
Spain	145
France	124
Mexico	110

```
In [117... plt.figure(figsize = (10,8))
country_plot = sns.barplot(x = df.country.value_counts()[:10].values,
                           y= df.country.value_counts()[:10].index,palette = "paste
for i in country_plot.containers:
    country_plot.bar_label(i)
```

/tmp/ipykernel_2979/434190962.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effect.

```
country_plot = sns.barplot(x = df.country.value_counts()[:10].values,
```



Rating

```
In [118... df.rating.unique()
```

```
Out[118... array(['PG-13', 'TV-MA', 'PG', 'TV-14', 'TV-PG', 'TV-Y', 'TV-Y7', 'R',
      'TV-G', 'G', 'NC-17', 'NR', 'TV-Y7-FV', 'UR'], dtype=object)
```

```
In [119... new_catgs = {
    'TV-PG': 'Parental Guidance',
    'TV-MA' : 'Mature Audience',
    'TV-Y7-FV': 'Teens',
    'TV-Y7': 'Teens',
    'TV-14': 'Teens',
    'R': 'Mature Audience',
    'TV-Y': 'General Audience',
    'NR': 'Mature Audience',
    'PG-13': 'Teens',
    'TV-G': 'General Audience',
    'PG': 'Teens',
    'G': 'General Audience',
    'UR': 'Mature Audience',
    'NC-17': 'Mature Audience'
}
df['rating']=df['rating'].replace(new_catgs)
df.head()
```

Out[119...

	show_id	type	title	director	cast	country	date_added	release_year	
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	No Cast	United States	September 25, 2021	2020	
1	s2	TV Show	Blood & Water	No Director	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	I Au
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Unknown	September 24, 2021	2021	I Au
3	s4	TV Show	Jailbirds New Orleans	No Director	No Cast	Unknown	September 24, 2021	2021	I Au
4	s5	TV Show	Kota Factory	No Director	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	I Au



In [120...

```
plt.figure(figsize= (10,6))
sns.countplot(x="rating", data=df, hue="rating")
plt.title("count of Rating by Movie and Shows")
plt.show()
```

